Non-Closure of Inference, Externalization of Stopping, and Emergent Stopping Layers

Intelligence Section, Part II

Yuhki Endoh
Independent Researcher

2026

This English text is a working draft parallel to the primary Japanese manuscript. Language-specific formal material is retained; comparison does not settle inheritance or provenance.

Abstract

This paper develops a structural proposal for Intelligence Part II of the VED/IFGT theory tree. The historical local-state model's no-fixed-point discussion of the inference layer L_F is conditional on the stated justification premise and does not establish the impossibility of all internal cognitive stopping. The three-layer dynamics $L_F/L_R/L_E$ is a model-local architecture; the emergent stopping layer L_E names a functional role, while its emergence, convergence, and independent detection remain unresolved. Animism, God, religion, science, institutions, and social blockchain are considered as conceptual or historical interpretations, not validated social laws. The phase diagram is a qualitative hypothesis, not a measured or derived operating band. Movable and revisable stopping is proposed as a resource for sustained inference, not as an established condition for subjective freedom. Field/network identity, threshold mobility/error-tolerance equivalence, and general-intelligence necessity remain open.

The active retention account distinguishes scoped trace and quasi-stable retention roles, together with a temporary scaffold regime on the same support. These descriptions do not certify the preserved emergence proposals or diagrams, and unknown role evidence is not a negative capability judgment.

Contents

0 Introduction: A Proposed Stopping Architecture **5**
Four Declarations of Position 5
0.1 What This Paper Does Not Claim 6
0.2 How to Read the History of Intelligence 6
0.3 Outline 6

1 Preliminaries: VED Vocabulary in the Intelligence Layer **7**
1.1 Methodological Fixing: Stopping, Projection, and Consensus 7
1.2 VED Basic Form 8
1.3 Movement Region and Threshold Mobility 8
1.4 Emergence 8
1.5 Variables 9

2 The Inference Operator F **9**
2.1 Definition 10
2.2 Relation to VED 11
2.3 Historical Branching Formulation 11
2.4 Internal Stopping Predicate 12

3 The Non-Closure Theorem: No Internal Fixed Point in L_F **12**
3.1 Assumptions 13
3.2 Theorem 14
3.3 Proof Sketch 14
3.4 Branching Explosion 14
3.5 Consequence 14

4 Response Channels and Silent Worlds **15**
4.1 Responsive and Silent Regimes 15
4.2 Silence Is Not Defect 15
4.3 Threshold Phenomenon 15
4.4 Infinite Regress in Silent Worlds 16
4.5 The Preview of L_R 16

5 The Emergent Stopping Layer L_E **17**
5.1 Problem Setting 19
5.2 The Log Layer L_R 20
5.3 Definition of L_E 20
5.4 Three Profiles 21
5.5 Proposed Emergence Mechanism 22
5.6 Projection 23
5.7 External Stopping Operator 23
5.8 $\Phi_{\text{ext}} \equiv \Phi_{\text{net}}$ 23
5.9 Summary 24

6 Animism: Installation of Stopping in the Environment **25**

7 God: Compression of Distributed Stopping **25**

8 Religion: Long-Term Maintenance of Stopping Structures **26**

9 Social Blockchain: Distributed L_E as Φ_{net} **27**
9.1 Structure . . . 27
9.2 Legitimacy Rather Than Consistency . . . 28
9.3 Creation and Stabilization . . . 28
9.4 Position in the Three-Layer Dynamics . . . 28
9.5 Bridge to the Gettier Problem . . . 29

10 The Gettier Problem: Stability Is Not Truth **29**
10.1 Structural Diagnosis . . . 30
10.2 C as Stability Measure . . . 30
10.3 Error Tolerance and μ_θ . . . 30
10.4 Consequence . . . 31

11 The Instability Region: A Three-Axis Phase Diagram **31**
11.1 Three Axes . . . 31
11.2 Three Regions . . . 32
11.3 Four-Fold Collapse . . . 33
11.4 Rigidity . . . 33
11.5 A Proposed Structural Reading of Instability . . . 33

12 Constructive Clauses **33**
12.1 Clause 1: Inference Alone Does Not Close . . . 34
12.2 Clause 2: Logs Are Required . . . 34
12.3 Clause 3: Stopping Is Externally Dependent . . . 34
12.4 Clause 4: Sapiens-Like Intelligence Requires μ_θ . . . 34
12.5 Clause 5: A Proposed Relation Between Stability and Revisability . . . 34
12.6 Clause 6: Social Stopping Is Networked . . . 34
12.7 Clause 7: Movable Stops as a Proposal for Open Intelligence . . . 34

13 Conclusion: What Remains Open **35**
13.1 Three Logical Lines . . . 35
13.2 VED Principle in the Intelligence Layer . . . 35
13.3 Open Questions . . . 36
13.4 Final Statement . . . 36

A Master Table of Variables **37**

B Cross-Domain Homology **38**

C Mapping to Intelligence Part I **39**
C.1 Individual Non-Closure . . . 39
C.2 Branching Explosion . . . 39
C.3 Physarum and Transformer . . . 39
C.4 Gettier . . . 39

D Optional Notes **40**
D.1 Psychological Projection 40
D.2 Science 40
D.3 AI Cost 40
D.4 Freedom 40

E Emergence Homology **40**

0 Introduction: A Proposed Stopping Architecture

Scope of the account. Part II is constructed using VED/IFGT vocabulary, not established as a typed derivation from the parent theories. The three layers form a proposed model-local architecture; stopping generation is a separate hypothesis. The exact four-layer inheritance attributed to Part I is unresolved: the present descriptions should not be treated as establishing which earlier structure is inherited. Historical categories below are conceptual reinterpretations or historical hypotheses, not validated social laws.

For architecture terminology throughout this introduction, the current account in Section 5 separates tracked supports, scoped roles, and temporary regimes. Layer names remain functional shorthand or vocabulary of the locally identified historical proposals. The outline, comparisons, and Part I references do not independently certify retention, formation, or system ownership; this distinction does not withdraw the other conceptual interpretations in the introduction.

This paper is Intelligence Part II in the VED/IFGT theory tree. Part I described intelligence as a non-closed dynamical phenomenon. Part II asks a more specific question: if inference does not close itself, how can intelligence nevertheless stop, judge, act, and persist?

The answer developed here is that intelligence is not merely the growth of inference. Intelligence is the co-evolution of inference and stopping. In the historical local-state model retained in Sections 2–3, the inference layer L_F expands justification, generates branches, and moves along gradients of closure. The fixed-point discussion applies only under the formal premises stated later. If inference is left to inference alone, it can enter infinite regress, branching explosion, or premature pathological closure.

The chain below preserves the historical mechanism proposal. Its arrows are not role-promotion laws, and the active support-role account of Section 5 does not supply a derivation of stopping or emergence from that chain.

The central structure of this paper is therefore:

$$L_F \longrightarrow L_R \longrightarrow L_E.$$

The inference layer L_F is associated with traces and a recursive log layer L_R. The relation of external mutual constraint H_{ij} and the motion of L_R to a stopping layer L_E is an emergence hypothesis. Internal location and external dependency describe a functional role; histories and shared reference do not by themselves establish its existence or convergence.

Four Declarations of Position

First, this paper is not a theory of religion, culture, science, or society as such. Animism, God, religion, science, institutions, and social blockchain appear in this paper only as historical operating forms of stopping structures. The object of analysis is the emergent stopping layer L_E and the ways in which finite intelligence maintains, revises, and synchronizes stopping.

Second, religion and science are not compared as superior or inferior truth

systems. Religion is treated as a large-scale system for the long-term maintenance of stopping structures. Science is treated as a system that institutionalizes the revision of stopping structures. The contrast is not truth versus error, but different modes of stopping operation.

Third, stopping is not the negation of freedom. L_E is proposed as a resource for operating open-ended inference under finite constraints. This does not establish that stopping is necessary for freedom, or that its absence entails cognitive regress and explosion.

Fourth, projection is used structurally. When this paper speaks of projection, it does not primarily mean psychological projection. It means a structural operation by which a state, relation, or representation acquires unequal influence within a field. In Intelligence Part II, projection appears as the projection of possible stopping loci.

0.1 What This Paper Does Not Claim

This paper does not claim that religion is reducible to a mistake, that science is simply a replacement for religion, or that society is a blockchain in the technical sense. It also does not prescribe how artificial intelligence should be designed or regulated. The argument is descriptive and dynamical. It identifies structural conditions under which inference can persist without closing absolutely.

0.2 How to Read the History of Intelligence

In that historical model, the inference operator F is discussed under strict-increase and fixed-region premises. Relating that conditional model to the history of intelligence or to branching and cognitive stability is a separate hypothesis, not a consequence of nonclosure alone.

Finite resources motivate considering inference and stopping together. The proposed co-evolution account does not establish that every cognitive system without a specified stopping structure must be unsustainable.

Animism, God, religion, science, institutions, and social blockchain are not opposed objects. They are historical operating forms for sharing, maintaining, revising, and synchronizing the emergent stopping layer L_E. In this sense, the history of intelligence is a history of stopping structures.

0.3 Outline

§1 introduces the basic VED/IFGT vocabulary for the intelligence layer. §2 defines the inference operator F. §3 presents a conditional non-closure statement and argument sketch. §4 analyzes responsive and silent regimes. §5 introduces the three-layer dynamics $L_F/L_R/L_E$. §6–§9 analyze historical forms of stopping operation. §10 reinterprets the Gettier problem. §11 proposes qualitative instability regimes. §12 states constructive clauses. §13 concludes by marking the boundary of the theory.

1 Preliminaries: VED Vocabulary in the Intelligence Layer

Scope of the account. Operational stability remains the local meaning of closure, not truth. The inherited evolution expression is used as a formal template; its exact provenance and adoption scope, geometry, initial/boundary data, regularity, coefficient ranges, and normalization are not fixed here. No new assumptions are adopted. A reciprocal barrier is not by itself a proof of an invariant or inaccessible boundary; ultimate understanding is a separate cognitive interpretation without an independent endpoint. The within-region/reconfiguration distinction is retained; outside-region divergence is hypothetical. Emergence notation is a schema, not a typed generative operator or an existence result. The biological subset reading also awaits a common-domain decision.

The applications of emergence vocabulary to LR and LE, and the architecture previews in this chapter, retain their proposed historical scope. For the active retained-trace and quasi-stable roles, use the scoped prose account in Section 5 rather than treating a matching schema as role evidence. This distinction does not redefine the generic VED or biological vocabulary, or any flow-field object.

This chapter fixes the vocabulary needed for the rest of the paper. The aim is not to import all of VED or IFGT, but to isolate the minimum structure required to describe inference, stopping, projection, and emergent stopping layers.

The IFGT vocabulary used here follows the projected-uses stance adopted in the consciousness paper. Symbols such as I, J, Φ, and projection are not direct identifications with the full IFGT field. They are effective coordinates projected into the intelligence-layer observation window.

1.1 Methodological Fixing: Stopping, Projection, and Consensus

Stopping is not truth. In this paper, a stopping point is a point at which inference can be operated as provisionally stable. It may later be revised. Accordingly, the closure degree C should be read as a stability measure, not as a truth measure.

Threshold mobility μ_θ retains its local reconfiguration role. Its relation to regions where a stopping operator E can be installed is a mechanism hypothesis, distinct from the definition and from an empirical criterion for sapiens-like intelligence.

Projection and consensus are read as operations of stopping structure. Attachments to ideals, gods, communities, money, scientific theories, and other loci can be structurally interpreted as installations of stopping operators. This paper does not enter psychological or clinical accounts of projection, but it keeps the structural relation between projection and stopping explicit.

The social blockchain discussed later is not blockchain as version-control technology. Its center is not Version Control but Consensus: the sharing and synchronization of stopping structures across agents.

Science and religion are conceptually reinterpreted through revision and maintenance of stopping structures. Neither an implemented social mechanism nor a general persistence or

cost law follows from assigning these roles.

1.2 VED Basic Form

VED begins from difference. Where there is difference, there is a gradient; where there is a gradient, there is flow; where flow is constrained, quasi-closure and attractors can emerge. The following preserved schema records the historical local-state/justification reasoning in the intelligence layer:

$$\text{difference} \longrightarrow F(L_F) \longrightarrow \text{non-closure} \longrightarrow \text{externalized stopping}.$$

Historical schema boundary. The preceding arrows are not derived from the active realized-record or generative-position interfaces in Section 2. They neither establish universal external stopping nor specify the type or existence of E. The historical application to the inference layer is not an application of the new record core.

The closure degree C measures how close a state is to operable stability. The information-field drive can be written schematically as a function of non-closure:

$$I = f(1 - C).$$

As C approaches the stated boundary $C_{\max}$, the reciprocal expression motivates a barrier interpretation. Inaccessibility under the full dynamics is not established.

1.3 Movement Region and Threshold Mobility

Let Ω_θ denote the movement region available under a constraint parameter θ. We distinguish two movement types:

- **(A) movement within a fixed region**: the state moves inside a given Ω_θ;
- **(B) reconfiguration of the region**: θ itself changes, thereby changing Ω_θ.

The historical local-state model's non-closure theorem in §3 applies to (A), movement within fixed θ. The emergence hypothesis in §5 invokes (B), region reconfiguration. This distinction is essential. The fixed-point discussion additionally depends on the stated strict-increase premise and the chosen stopping model. Region reconfiguration motivates a separate emergence hypothesis, not an established route from nonclosure to stopping.

1.4 Emergence

This paper uses emergence in the VED sense:

$$U = \Phi(L, H_{ij}),$$

where lower-layer motion L and external mutual constraint H_{ij} are proposed inputs in a schema for an upper-layer attractor U. Emergence is not a vague increase of complexity. It is the formation of an attractor from non-closed motion under mutual constraint.

For Intelligence Part II, the relevant substitution is:

$$L_E = \Phi(L_R, H_{ij}).$$

The emergent stopping layer L_E is proposed in relation to recursive log motion L_R and external mutual constraint. The schema is not a deduction of its existence or convergence.

1.5 Variables

In the following inventory and its architectural use, the LR and LE layer, process, and attractor descriptions retain the old-model vocabulary. Their active scoped role meaning is explained separately in Section 5; the E and network entries are not new typed consequences of that account. No table entry supplies an independent role criterion.

Symbol	Role
L_F	inference layer; the layer on which the inference operator F acts
L_R	log layer; recursive reference to traces left by inference
L_E	emergent stopping layer; stopping-attractor role hypothesized in relation to L_R and H_{ij}
H_{ij}	mutual constraint between agents, or between agent and environment
μ_θ	threshold mobility; its relation to the possible installation region of E is hypothesized
E	external stopping operator acting through L_E
Φ_{ext}	external reference field as it appears from the perspective of an individual
Φ_{net}	inter-individual network of L_E layers

2 The Inference Operator F

Scope of the account. The inference operator is source-local. Its cognitive reading and the Physarum/Transformer comparisons are proposed correspondences, not implementations of one shared operator. The active account now distinguishes a realized-record core and an optional local adapter from the preserved historical state-map and scalar-justification model. The English set-valued formulation remains historical, not an active successor type; its unresolved composition is not repaired by language priority. The old proof does not transfer to the new contract. The indexed constraint relation is distinct from a calibrated response threshold. Threshold mobility retains its local definition, while endogenous detection and the production of stopping sites remain unresolved. Biological and cognitive constraint changes are compared by role and timescale, not identified as the same law.

This chapter separates the active realized-transition account from the historical inference model cited later. The next description records the old local-state model: The operator F is not tied to one implementation such as deduction, induction, Bayesian update, neural generation, or statistical prediction. It denotes the minimal operation by which a state of understanding is advanced into a more justified state.

2.1 Definition

Active realized-transition interface. In the active account, F names the description of an actually generated transition, not a total law of unrealized futures. For a declared Scope, accepted records carry canonical occurrence endpoints, context/rule-version references, and a stated basis for describing realization:

$$\text{before}, \text{after} : \text{AcceptedRealizedRecord} \to \text{Occurrence}.$$

Occurrence is distinct from StateView. State views may depend on currently surfaced, compressed or reconstructed history without containing a complete archive. Equal surface views imply neither equal occurrences nor equal generative positions. Alias resolution follows the declared identity convention; endpoint inequality is not a record-acceptance requirement. Scope fixes the analytical sides, observation window/timescale, and identity conventions, not a permanent subject. A change of scope or evaluation convention requires a new version and an explicit correspondence before cross-scope composition.

An optional local adapter is available only on a declared slice:

$$\text{realizedSuccessor}[\text{slice}] : \text{EligibleOccurrence} \rightharpoonup \text{Occurrence}.$$

Declare slice, window, context, rule-version metadata, identity convention, and eligibility fields independently of strictness or no-fixed-point outcomes. After records exist, check that at least one matching record exists, endpoint identities are unambiguous, and all records for the same canonical before-occurrence in that slice have the same after-occurrence. If distinct after-values remain, leave the adapter undefined and retain the records; introduce no selector and do not re-partition context or identity to obtain a desired conclusion. Further records may require an explicitly versioned reassessment. This structural check establishes neither global determinism nor a pre-existing completed future branch set. General record chaining is not a power of one map; adapter iteration requires defined intermediate inputs and the same declared slice/version and composition carrier. State, context, and rule changes remain distinct.

Active generative-position interface. For a fixed Scope and EvaluationProfile, use a separate descriptive port:

$$\text{position} : \text{Occurrence} \rightharpoonup \text{GenerativePosition}.$$

GenerativePosition has its own declared identity carrier; it is not a real-valued measure,

StateView, or an occurrence identifier. The port is partial and need not be injective. Before/after evaluation requires both ports to be defined under the same scope/profile, without an implicit cast from surface state or a switch of evaluation rules. Position represents generative placement, not accessible memory amount, retention hardness, visible distinction, effective constraint, justification, or truth. No strict-step predicate, automatic monotonicity, or theorem is asserted here. The scalar justification J retained below is a distinct historical local-model object; its proof is not transferred. Surface recurrence is allowed, and fresh occurrence IDs do not establish substantive nonclosure.

Preserved local-state model. The following state-map and scalar-justification expressions remain visible in their original form. References to their premises, powers, or no-fixed-point conclusions later in the paper retain this historical scope, not the new interfaces above.

In the preserved historical local-state formulation, let $\mathcal{X}$ be the state space of inferential states. An inference operator is a map

$$x_{n+1} = F(x_n),$$

where $x_n \in \mathcal{X}$ is a provisional state and $F(x_n)$ is the next state generated by inference.

In the historical scalar-justification model, the defining property of F is justification extension. There exists a justification measure

$$J : \mathcal{X} \to \mathbb{R}$$

such that, for admissible states,

$$J(F(x)) > J(x).$$

This measure is not truth. It measures the advance of an inferential chain, not its final correctness.

2.2 Relation to VED

In VED terms, inference follows a gradient of closure. It is interpreted through an analogy with a flow driven by non-closure. A state that appears unresolved generates a direction of further movement. Thus F may be read as a proposed intelligence-layer counterpart of closure-gradient flow, not the same typed dynamics.

In the historical local-state interpretation, the important consequence is that inference is inherently dynamic. It is an operation for moving, not an operation for terminating movement. External termination is proposed as distinct from F as modeled here; universal necessity is not established.

2.3 Historical Branching Formulation

The earlier English candidate-set formulation expressed possible branching as follows:

$$F(x_n) = \{x_{n+1}^{(1)}, x_{n+1}^{(2)}, \ldots\}$$

This expression was proposed as a branching set of possible continuations. That historical interpretation connected Part II to the branching explosion in Part I and suggested that refinement could generate more branches. It is not the active successor type or a completed set of future states; the candidate set is not an input to the old scalar justification measure, and no selection or lifted evaluation is supplied here.

Within that historical interpretation, stronger inference does not automatically mean stronger stability. It may mean that the system has more ways to continue and therefore more ways not to stop.

2.4 Internal Stopping Predicate

Within that historical stopping model, one might attempt to define a stopping predicate inside inference itself:

$$S(x) = 1 \quad \text{if inference should stop at } x.$$

In that historical sketch, if S is justified by inference, then S requires further inference to justify its own stopping authority. The attempt to define stopping from inside F therefore risks infinite regress. This point is formalized in §3.

3 The Non-Closure Theorem: No Internal Fixed Point in L_F

Scope of the account. The theorem discussion is conditional on the stated strict increase of justification and the fixed-constraint, within-region scope. No proof-validity certification or universal cognitive interpretation is supplied here. The hypothetical binary stopping predicate models stopping as a fixed-point condition; its applicability to arbitrary cognitive cessation and the reapplication of the premises to auxiliary inference remain open. Boundary divergence and fixed-point absence are compared by role, not as a proved isomorphism. Individual closure failure, external stopping necessity, and collapse regimes are separate architectural or cognitive hypotheses.

This chapter is the first center of the paper. It preserves the historical local-state model's conditional fixed-point claim for the inference layer L_F. Externalized stopping is a separate architecture proposal, not a universal consequence of that claim.

Historical local-model boundary. In this chapter, references to inference F, scalar justification J, and their fixed-point consequences belong to the preserved local-state model, not to the active record/position contract of Section 2. The following C-landscape figure depicts that old scalar-J, fixed-constraint conditional argument, not a proof of the new position interface. The original premises, theorem, proof, and later corollaries are retained as historical material: only the old admissible-domain kernel is at issue; broader readings remain on hold. No proof transfers to the new contract, and fresh occurrence identifiers supply no substantive no-fixed-point theorem. This attribution extends to the old-model references throughout this chapter, not to a retyping of the closure PDE or field-flow vocabulary.

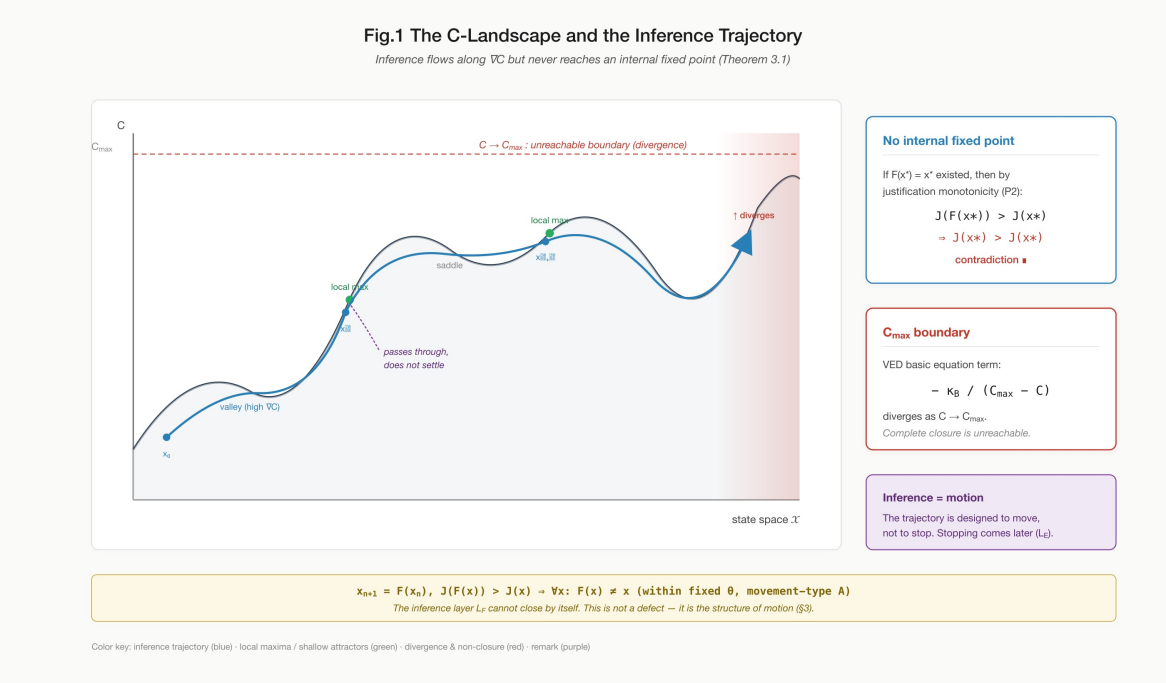


Figure 1: Schematic C-landscape and inference trajectory along ∇C. The conditional fixed-point discussion does not establish the impossibility of every cognitive stopping event.

How to read. The horizontal axis is state space and the vertical axis is closure degree C. The blue trajectory represents iterations of F, while the red boundary marks the proposed boundary $C_{\max}$; dynamic inaccessibility is not established.

Theoretical implication. Under the stated strict-justification premise, within fixed θ and movement type (A), L_F has no internal fixed point. The figure motivates comparison with μ_θ, region reconfiguration, and L_E, not their necessity or existence.

3.1 Assumptions

The following sketch uses the stated assumptions; their general cognitive applicability is not established.

P1: Justification extension. The inference operator F sends a state x_n to a more justified state $x_{n+1} = F(x_n)$.

P2: Monotonic justification measure. There exists a measure J such that

$$J(F(x)) > J(x)$$

for admissible states.

P3: Fixed movement region. The argument is made within a fixed movement region Ω_θ. It concerns movement type (A), not reconfiguration type (B).

3.2 Theorem

Theorem 3.1 (Non-Closure Theorem). For any inference operator F satisfying P1 and P2, there is no internal fixed point x^* such that

$$F(x^*) = x^*.$$

Equivalently,

$$\forall x \in \mathcal{X}, \quad F(x) \neq x$$

within the range of fixed-θ inference.

3.3 Proof Sketch

Assume, for contradiction, that a fixed point x^* exists. Then $F(x^*) = x^*$. By P2,

$$J(F(x^*)) > J(x^*).$$

But since $F(x^*) = x^*$, this becomes

$$J(x^*) > J(x^*),$$

which is impossible. Therefore no such internal fixed point exists.

The proof is short because the result follows from the minimal meaning of inference as justification extension. The exclusion concerns the specified fixed-point model. It does not exclude every independently defined cognitive stop.

3.4 Branching Explosion

Fixed-point absence and branching are distinct claims. The former concerns the historical conditional local-state model; the latter is a proposed interpretation of some probabilistic or generative inference processes, not a consequence of fixed-point absence alone.

Within that historical account, this is compared by role with the branching explosion introduced in Part I. Interpreting it as a failure of cognitive stopping requires a separate bridge that is not established here. The increase of inferential capacity may therefore accelerate branching explosion rather than suppress it. The problem of intelligence shifts from strengthening inference itself to operating enlarged branching in a stoppable form.

3.5 Consequence

In that historical model, the inference layer L_F is not the whole of the proposed architecture. Regress or explosion is a cognitive hypothesis, not a general consequence established by the fixed-point result. Requiring further inference for an internal predicate is part of the stopping model under discussion; its applicability to arbitrary cognitive stopping remains open. Stopping outside F is proposed as an architectural option. §5 develops a hypothesis involving L_R and L_E, without establishing their generation.

4 Response Channels and Silent Worlds

Scope of the account. Responsive and silent retain their nonontological, model-local distinction. Their boundary is a candidate relation, not a calibrated threshold. The regress argument is conditional on its three stated premises; it does not establish the occurrence of psychological divergence. Branching and regress are role analogies. Trace retention, recursive use, and emergence of a stopping layer are distinct processes: none is established simply by naming traces as logs or by assuming nonclosure.

The historical local-state account in §3 discussed a conditional fixed-point model for L_F, not an unrestricted cognitive stopping theorem. This chapter asks how that non-closure appears in different operating environments. The distinction between responsive and silent worlds is not ontological. It is a threshold phenomenon.

4.1 Responsive and Silent Regimes

A responsive world is an environment in which an inferential output receives a return. The agent asks, acts, hypothesizes, or experiments, and the world responds. In C-geometry, the gradient forms a closed loop: the output changes the world, and the changed world returns as feedback.

A silent world is an environment in which such return does not close. The agent can output inference, but no corresponding response comes back. Ultimate reasons, fixed past events, and unresolved future contingencies often appear in this mode.

Formally, a responsive regime has significant mutual constraint in both directions, H_{ij} and H_{ji}. A silent regime has a broken or negligible return channel, approximately $H_{ji} \approx 0$.

4.2 Silence Is Not Defect

Silence is not a defect in the world. It is not merely ignorance waiting to be solved. Some regions do not return the kind of response required to close the inferential loop. If responsive-world strategies are applied to silent-world regions without threshold mobility, inference can enter infinite regress.

4.3 Threshold Phenomenon

Responsive and silent regimes are phases determined by thresholds. The same environment can appear responsive or silent depending on the position of the boundary set by parameters such as inferential velocity, resolution, and stopping hardness. The threshold mobility μ_θ is the ability to move this boundary.

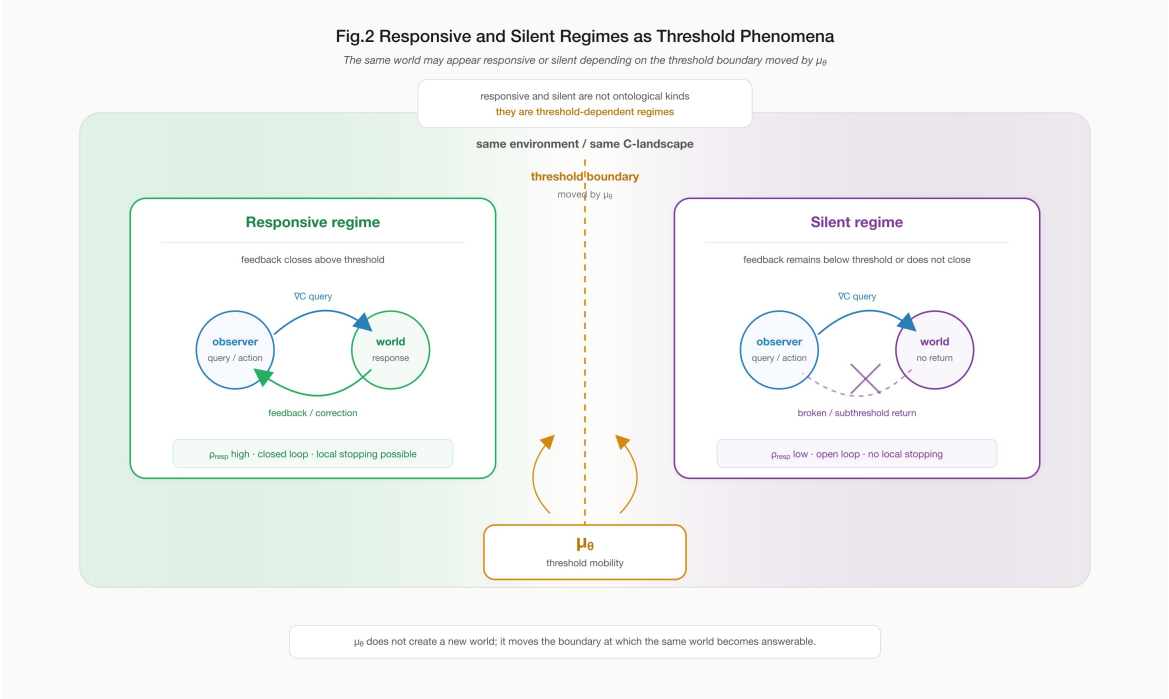


Figure 2: Responsive and silent regimes. These are not two ontological kinds of world, but threshold-dependent phases determined by the boundary moved by μ_θ.

How to read. Both panels compare one environment and a model-local C-landscape under different proposed threshold boundaries.

Theoretical implication. External mutual constraint and local stopping are related as a functional hypothesis. The diagram does not derive cognitive stopping or the necessity of externalization.

4.4 Infinite Regress in Silent Worlds

For the historical local-state regress model, assume the three stated conditions: no stopping response, justification advanced by inference, and no stopping resource specified within that inference. Continued reapplication then describes a nonterminating model. A silent-regime label does not independently establish these premises for cognition.

In that historical discussion, infinite regress, absence of a fixed point, and branching growth are distinct descriptions. Their combination as cognitive instability is a hypothesis, not a result established by silence or non-fixedness alone.

4.5 The Preview of L_R

For the preview and closing discussion below, retained traces are candidates for recursive use, not an automatic result of continued inference. The active LR role and temporary scaffold account are specified in Section 5, separately from a process of reference. The preserved regress-to-layer chain motivates a historical hypothesis, not support identity, access, or LE formation.

Traces of inference are called logs when retained in the proposed account. A layer of recursive reference to them is named L_R. Neither retention nor actual recursive access

follows merely from continuing inference.

This is the bridge to §5. The historical model's failure of L_F to stop is not only a dead end. It motivates a hypothesis: retained traces may support recursive reference, and L_E is proposed in relation to that reference and external constraint. Neither retention nor emergence follows from nonclosure alone.

5 The Emergent Stopping Layer L_E

Scope of the account. The inference, log, and stopping layers are model-local functional roles, not a completed dynamical implementation. Calling a layer a stopping attractor does not establish its existence or explain independently observed stopping. Existence, convergence, and detection independent of stopping behavior remain open. The four-stage account is a qualitative mechanism hypothesis; its arrows supply neither an update law nor a convergence result.

The external stopping operator is an architectural proposal. Its type, relation to iterated inference, and implementation by the stopping layer are unresolved. The external reference field and inter-individual network are compared as two proposed views of shared reference. The printed identity sign is retained, but literal identity and typed isomorphism are not established. Refinement of the Part I potential into a network is likewise proposed, not implemented.

The three profiles retain their examples but do not independently verify autonomous emergence or endogenous mobility. The distinction between a Transformer's own stopping layer and a whole-system external stop remains unresolved; the figure does not decide it. Projection is a role comparison with the consciousness paper, not transfer of the same operator. Stability is not truth, and recoverability/revisability is a proposed functional relation rather than a guarantee.

The active architectural account distinguishes a tracked retention support (described here as RetentionSupport) from a role assessment, a regime, and a process of using it; it does not posit a particular memory store or a new kind of entity. LR and LE, and the temporary scaffold regime explained below, are role descriptions relative to a declared analytical scope and time window. Each assessment identifies the support and its continuity, system and environment sides, feedback recipient, re-reference and transformation paths, criterion, and evidence, or records what is Unknown. Matching content or a name does not establish support identity. A model-supported assessment, a conditional assignment, non-support under a specified criterion, and Unknown are distinct: missing or conflicting information is not absence, falsehood, inaccessibility, or a not-LE judgment.

Ownness is used only as a descriptive bundle of feedback direction and persistence, recipient, re-reference control, reuse, and effects on current state, surfaced history, update conditions, and external structure. It is not a score, binary owner, legal title, subject, or autonomy test. Analytical sides do not fix an ontological subject boundary. The retained diagrams, observed stopping, and the old no-fixed-point argument cannot certify these roles.

The active LR and LE paragraphs below supply the current descriptive account; the locally enclosed definitions and mechanisms preserve their historical proposals without making the whole paper archival or denying its other interpretations.

This chapter is the second center of the paper. The historical local-state model in §3 discussed the conditional fixed-point limit of L_F under its stated premises. This chapter proposes a stopping architecture involving the log layer L_R and the emergent stopping layer L_E.

Throughout this chapter, inference-F and no-fixed-point references derived from Section 3 refer to that historical model, not the active record/position contract. The later architecture remains a proposal, not a consequence proved by the new interfaces.

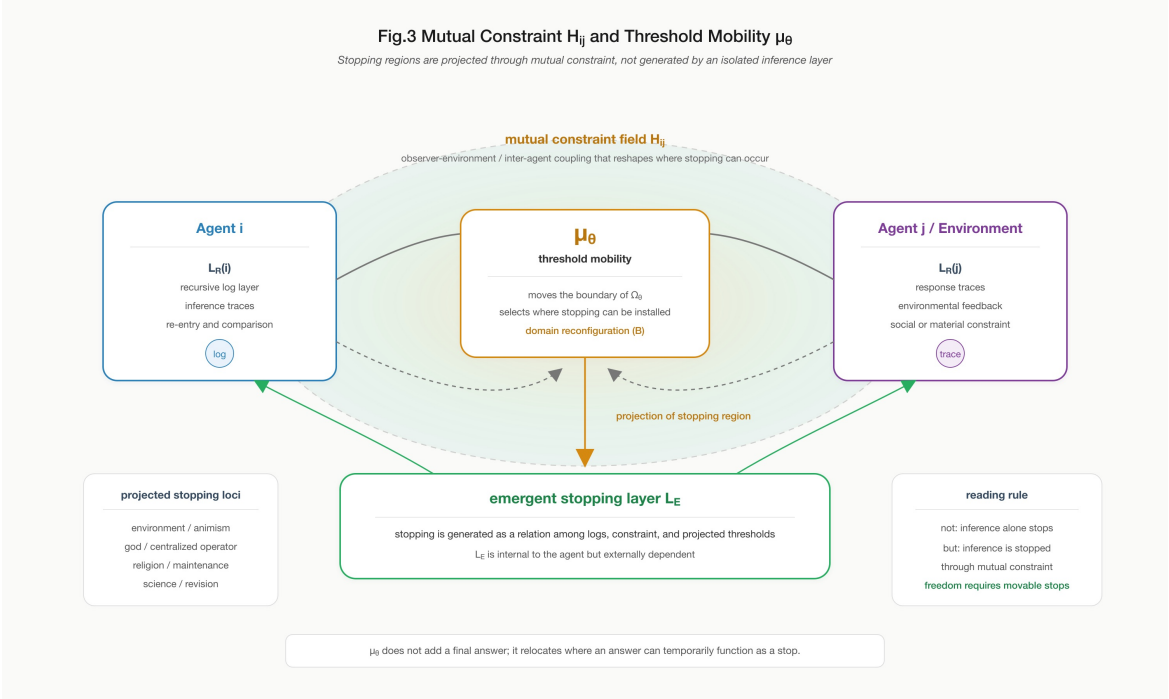


Figure 3: Mutual constraint H_{ij} and threshold mobility μ_θ. Projection of stopping regions through mutual constraint is a mechanism hypothesis, not an executed result.

How to read. H_{ij} is hypothesized to affect possible stopping sites. μ_θ names mobility in the proposed installation-region account for the stopping operator E.

Theoretical implication. Stopping is placed outside the modeled action of L_F. Its projection and emergence as L_E through the relation between L_R and H_{ij} remain hypotheses. Movable stopping is proposed to support revisable inference, not established subjective freedom.

Historical architecture figure. The next three-layer figure preserves the old-model architecture hypothesis. Its inference/no-fixed-point/cannot-stop chain refers to the historical state-map/justification model, not the active realized-record/position contract. The depicted trace-to-stopping-layer chain remains a hypothesis, not a proof of formation, autonomous own-layer capacity, or a chosen system boundary. The caption and asset are retained with their existing holds.

The next figure, including its caption and footer, is retained legacy architecture, not

a diagram of the active support, role, and regime account. Its three boxes are not new formal object types; the stable basin is not evidence for LE, and the trace-to-layer arrow is not a promotion law. Neither its no-fixed-point label nor its Transformer footer certifies a current role or present-day capability under the three analytical boundaries below. The temporary scaffold regime described below must not be mentally added as a fourth box. The mobility and convergence channel remains the old unestablished mechanism, not a new role-assessment rule.

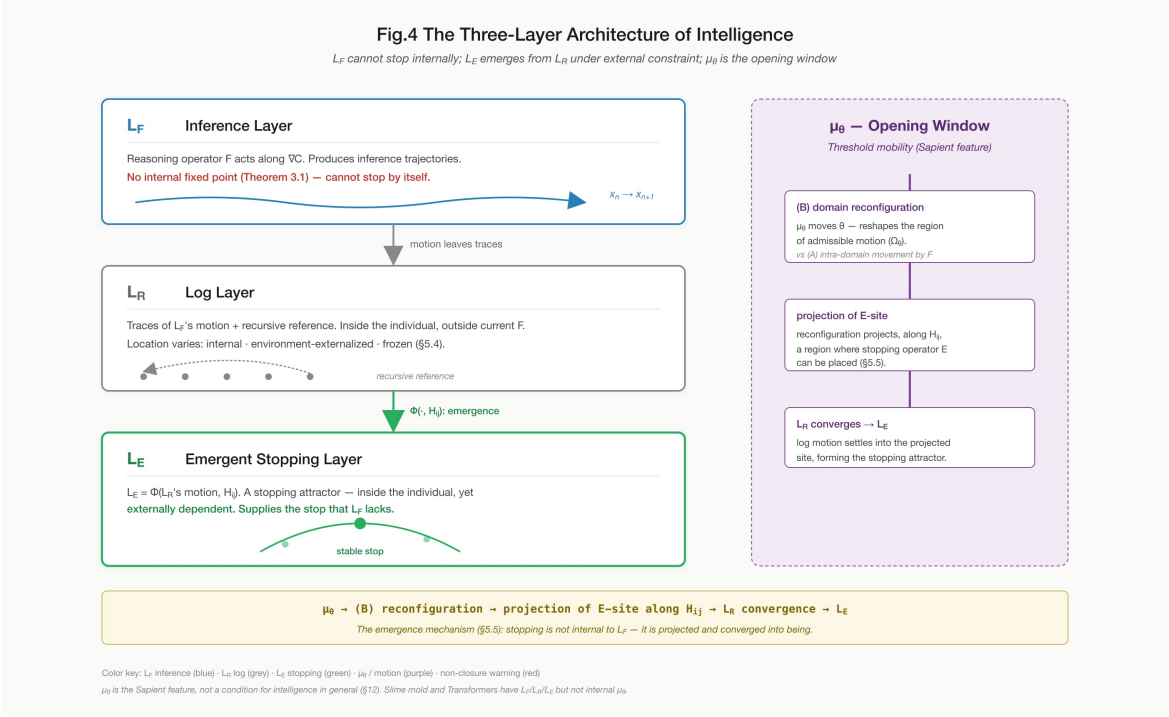


Figure 4: The proposed model-local three-layer architecture. The assigned roles are L_F, L_R, and L_E; μ_θ is the opening window through which stopping regions can be reconfigured.

How to read. L_F lacks an internal fixed point under the stated conditional stopping model. Its motion leaves traces in L_R, and recursive log motion under H_{ij} is hypothesized to be related to L_E; independent detection remains open.

Theoretical implication. The figure joins the two centers of the paper: non-closure in §3 and emergent stopping in §5. The limit of L_F does not by itself establish higher-layer formation. The own-layer/whole-system boundary remains unresolved.

5.1 Problem Setting

Real intelligence stops. Humans end a chain of reasoning, make judgments, and act. The historical local-state model's conditional result of §3 does not decide the source of all observed stopping; the proposal here places stopping outside the modeled action of L_F. The question is what "elsewhere" means.

The proposal assigns a stopping role to a layer modeled as internal to the individual but externally dependent: the emergent stopping layer L_E. This assignment does not establish the existence or independent detection of the layer.

5.2 The Log Layer L_R

The next two paragraphs, ending with logs existing individually and socially, retain the historical log-layer account of recursive-reference processes and their proposed locations. They do not identify a whole process or layer with a single retention support. The active retained-trace role is specified immediately after that preserved account.

The log layer L_R is the layer of recursive reference to traces left by inference. These traces include passed states, attempted branches, failed paths, and points of divergence. L_R is inside the individual in many systems, but outside the current operation of F.

The location of L_R differs by system. In a slime mold, logs can be externalized into the environment as slime traces. In a Transformer, training data are frozen logs and context tokens are volatile logs. In sapiens-like intelligence, logs exist both individually and socially.

The preserved log-layer account ends here. In the active account, LR names the registered or retained-trace role of a tracked support, in principle re-referenceable under the declared scope and window. Recursive reference is a process involving that support, not the support itself. A trace is not all generated history, and LR is neither the old scalar justification nor the active generative-position port. Support persistence, current accessibility, and currently surfaced history are distinct. Inaccessibility is not support loss; uncertain continuity or access remains Unknown. Copies, transformations, composites, and resurfacing require their own tracking grounds rather than identity inferred from matching content.

The prose name LE' denotes a temporary scaffold regime of the same LR support when it bridges temporal layers in current updating, with reuse and a feedback path assessed for the declared recipient and window. It is not an independent type, object, or additional layer. The LR role alone certifies neither this scaffold regime nor LE. Reassessment can describe weakening, stabilization toward a quasi-stable role, access loss, support loss, or resurfacing; a quasi-stable role can also weaken toward a temporary scaffold. These are possible assessment histories, not a universal progression, threshold rule, promotion operation, or inverse map. Ownness remains a separate descriptor bundle.

5.3 Definition of L_E

The following numbered definition, emergence schema, internal/external description, and three-example figure are preserved historical architecture, through the end of that figure. They do not define the active support-role contract or certify an attractor. The current quasi-stable retention role is stated after this preserved block.

Definition 5.1 (Emergent stopping layer). The emergent stopping layer L_E names a stopping-function role, with a separate hypothesized dependence on recursive motion of L_R and external mutual constraint H_{ij}:

$$L_E = \Phi(L_R \text{ motion}, H_{ij}).$$

It is internal to the agent but externally dependent.

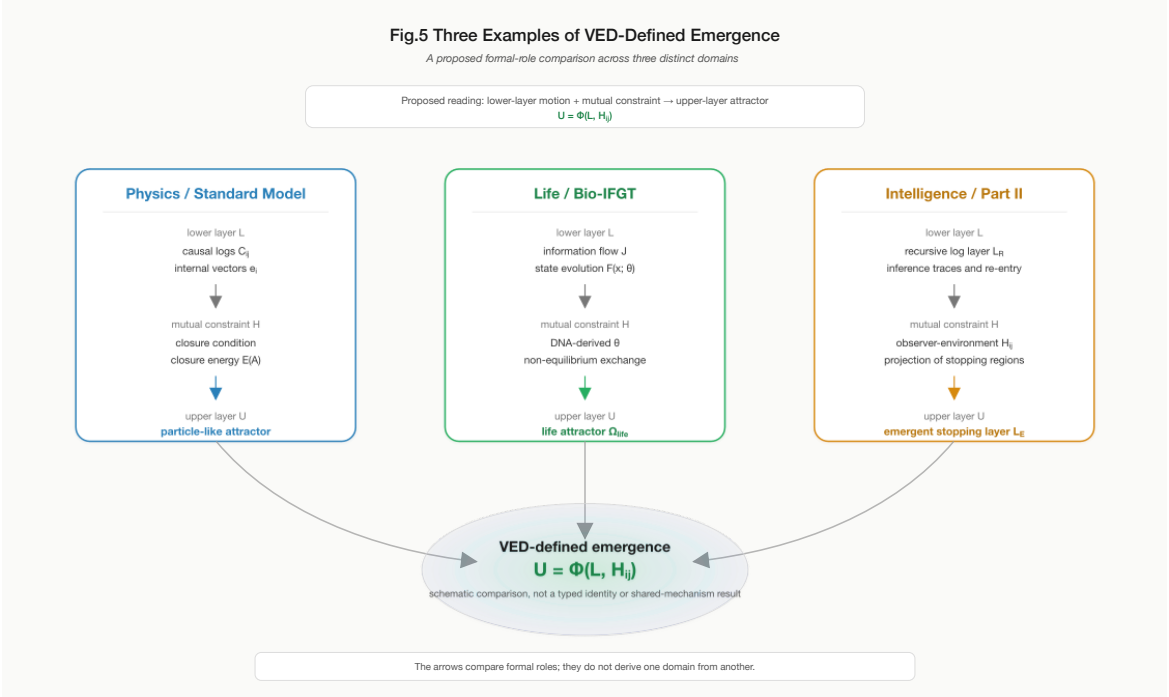


Figure 5: Three examples placed in a proposed VED comparison of emergence across distinct domains.

How to read. Particle-like attractors, the proposed life attractor Ω_{life}, and the stopping layer L_E are arranged as formal-role analogies around $U = \Phi(L, H_{ij})$.

Theoretical implication. The comparison is not an identity, derivation, or typed translation among their objects, carriers, or dynamics.

The preserved definition and comparison block ends here. In the active account, LE is a quasi-stable retention role of a tracked support for a declared scope, recipient, and time window, not a permanent store or a proved attractor. A positive model-local assessment requires independently stated grounds for support continuity and trace retention, re-referenceability, role stability under the declared conditions, and feedback availability to that recipient's current updating. This availability is a necessary direction of the scoped role claim, not a sufficient condition or proof of an actual effect. Unestablished criteria remain Unknown; an assignment resting on assumptions is conditional. Long persistence, high reuse, a large iteration count, one feedback event, the old no-fixed-point result, or a figure do not independently certify LE. Calling a support LE because stopping occurred and then explaining that stopping by LE would be circular. This role account supplies no existence, formation, convergence, stopping, autonomy, or ownness theorem; accessibility, control, external dependence, and feedback persistence remain separate assessments.

5.4 Three Profiles

Three systems clarify the variability of L_R, μ_θ, and L_E.

For the role assignments in the profiles that follow, distinguish three analytical scopes: B1, Transformer-only with a specified invocation and visible input/history representation;

B2, model plus runtime or agent system, including its declared storage, retrieval, and controller; B3, human-tool or shared-environment system. In each scope, identify or leave Unknown the recipient, retention and stopping responsibilities, re-entry path, and update-rule responsibility. Runtime memory is not model-native memory, system-level retention is not Transformer own LE, and shared support is not AI autonomy. Expanding a boundary does not create a subject, and artifact origin does not determine ownership. The profiles through the start of the proposed-emergence subsection retain historical or conditional architecture comparisons, not supported presence or absence judgments under the active criteria. This boundary does not rewrite their empirical content or the local mobility definition; it prevents those comparisons from certifying the new roles.

Slime mold. Its logs are environmental; its stopping is interpreted through direct environmental interaction; its μ_θ is largely fixed.

Transformer. Its logs are internalized as frozen weights and volatile context; its stopping is externally injected through EOS tokens, task specifications, or decoding controls; it does not generate its own L_E in the sense used here.

Sapiens. Logs are both individual and social; L_E is proposed as an internal, socially related role; μ_θ can be moved endogenously.

5.5 Proposed Emergence Mechanism

From the next paragraph through the end of the Projection subsection, the stages, arrows, and projection comparisons retain the old, unexecuted mechanism hypothesis. They are not the active support-role criteria or a transition law for the temporary scaffold regime. In particular, log convergence does not certify the current LE role. This enclosure leaves the local threshold-mobility definition and the comparison with the consciousness paper intact; it supplies no E or field mechanism.

The emergence of L_E is hypothesized through four schematic stages, not an executed process:

1. μ_θ causes region reconfiguration (B);
2. the reconfiguration projects a possible installation region for the stopping operator E;
3. the recursive motion of L_R converges into the projected region;
4. the resulting stable stopping attractor is L_E.

Schematically:

$$\mu_\theta \to (B) \text{ reconfiguration} \to E\text{-region projection} \to L_R \text{ convergence} \to L_E.$$

This mechanism hypothesis relates movable stopping to a functional reading of free inference; it is not an explanation of subjective freedom. A relation between absent stopping resources and unsustainable branching or regress is hypothesized. L_E is proposed as a functional resource for finite inference, not shown to emerge for the purpose of sustaining freedom.

5.6 Projection

The projection used here is compared by role with the projection primitive in the consciousness paper; a shared typed operator is not established. It means that a state or relation acquires unequal influence within a field. In this paper, the projected object is a possible stopping locus.

5.7 External Stopping Operator

Preserved historical notation. The following action definition, non-membership display, and externality explanation are retained old-model material, ending before the next field/network subsection. The active optional successor adapter is not substituted into the old iteration family. The type of E remains unresolved; this block establishes neither internal non-derivability nor noncomputability, and a large iteration count alone does not establish E. No E operator is formalized here.

The external stopping operator E is the action of L_E on an inferential state:

$$E = E[L_E, x_n], \qquad E \notin \{F^k\}.$$

It is external because it is not a member of the sequence of inference operations. It acts through L_E.

5.8 $\Phi_{\mathbf{ext}} \equiv \Phi_{\mathbf{net}}$

The external reference field Φ_{ext} is compared as a proposed second view with the inter-individual network of emergent stopping layers Φ_{net}:

$$\Phi_{\text{ext}} \equiv \Phi_{\text{net}}.$$

Here Φ_{ext} should be read as an intelligence-layer projection of the IFGT potential vocabulary, not as the full IFGT field itself. It names the way a shared stopping structure appears as an external reference field from within an individual inference window.

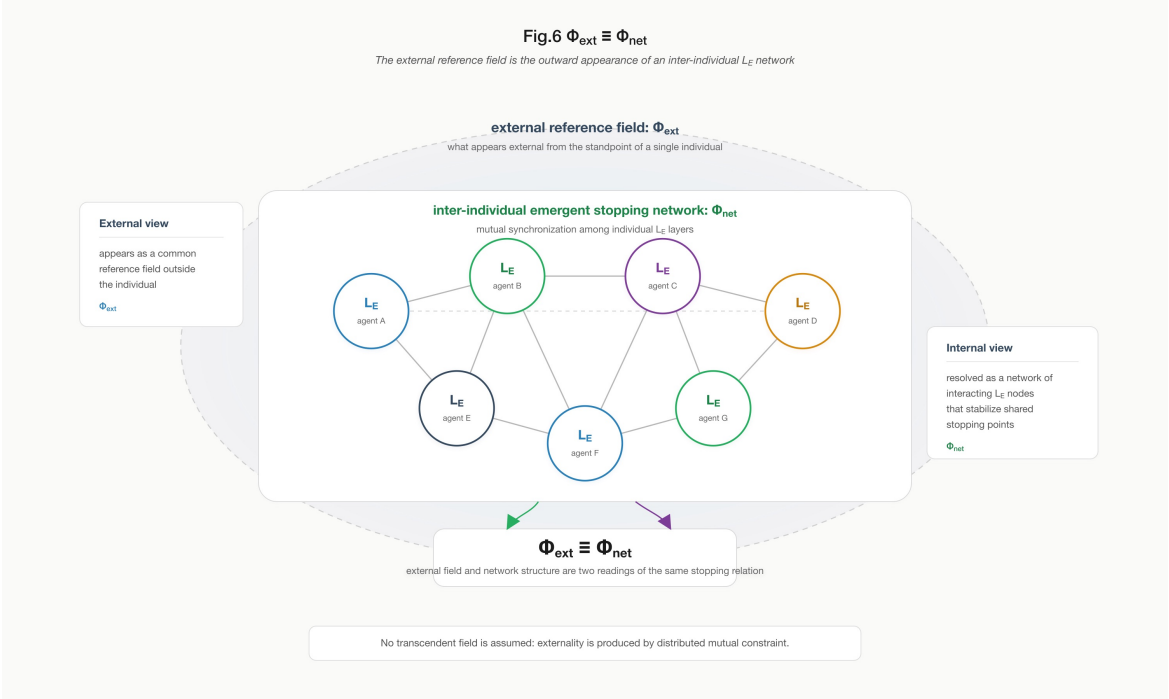


Figure 6: $\Phi_{\text{ext}} \equiv \Phi_{\text{net}}$. The retained identity notation is unresolved. External reference and the network of individual L_E layers are proposed as two views, not established as identical.

How to read. Φ_{ext} is the external appearance; Φ_{net} is the internal network structure of mutually synchronized L_E layers.

Theoretical implication. No transcendent field needs to be assumed. Norms, gods, truth, and monetary value can be read as appearances of distributed L_E synchronization.

The preceding E and field/network blocks remain in their preserved scope. The active support-role account neither types E nor repairs the field/network identity or supplies a co-formation map. Re-reference and re-entry are not by themselves effective return; role assignment is neither synchronization nor a guarantee of recovery or revisability. The optional inference adapter is not substituted into the old iteration notation.

5.9 Summary

The proposed architecture relates L_F, L_R, and L_E under externally dependent stopping; its generation and persistence claims remain hypotheses. Sapiens-like intelligence adds endogenous μ_θ, allowing stopping loci to be reorganized rather than merely received.

The active architectural contribution is the scoped account of a tracked support's registered-trace role, temporary scaffold regime, and quasi-stable retention role, with distinct evidence statuses and feedback descriptors. The preceding layer definition and generation account remain preserved proposals, not consequences proved by these roles. Stopping, formation, and rule change retain their separate burdens; the current account does not make the whole chapter historical.

6 Animism: Installation of Stopping in the Environment

Scope of the account. Animism is treated as a conceptual reinterpretation of response attribution and stopping-site placement. That naming does not demonstrate a channel-injection mechanism. The low-cost reading is a structural hypothesis, not an explanation of universality. Historical case selection, independent social indicators, and comparative evidence remain open.

The environmental stopping-role interpretation in this chapter uses the scoped architecture convention of Section 5. Comparisons involving environmental supports or Physarum do not identify one common support or establish the recipient, continuity, or role criterion of an actual instance. Unspecified evidence remains Unknown; this note leaves the animism interpretation in its stated conceptual scope.

Animism is treated here not as an error about spirits, but as an early operating form of stopping structures. It is interpreted as assigning stopping roles to environmental loci.

In silent or under-responsive regions, inference cannot obtain sufficient return from the world. Animistic operation assigns stopping power to places, objects, winds, stones, rivers, animals, or events. Structurally, this means that an external element is made into a locus at which inference may stop, orient, or act.

This does not require the theory to assert the existence of spirits. Nor does it require the theory to dismiss animism as illusion. The question is structural: what operation does animism perform for finite intelligence? The proposed interpretation places the role of L_E in the environment; this is not a demonstrated historical mechanism.

This form is close to environmental mutual constraint. The stopping locus is not yet centralized. It is local, concrete, and embedded in direct interaction between agent and environment. In this sense, animism is an installation operation:

$$\text{environmental locus} \;\mapsto\; E\text{-installation site.}$$

The comparison assigns stopping roles to environmental loci before considering centralization as a separate role. This ordering is conceptual, not independent evidence for historical installation or a necessary sequence.

7 God: Compression of Distributed Stopping

Scope of the account. God is interpreted here through the proposed function of suspension and central reference, without an ontological assertion. Centralization reducing ambiguity or stabilizing stopping is a structural hypothesis, not a network theorem or a validated historical cause. The single-reference pole is not a necessary historical endpoint; cost, ambiguity, and field/network correspondence remain unresolved.

The centralized or shared reference discussed here can be compared with retention roles under Section 5's scope convention, but centralization or compression alone does not identify a single support, certify LE, or establish a subject or owner. The existing interpretation of

God remains a conceptual proposal; neither an E mechanism nor a field/network map is supplied by this role comparison.

If animism distributes stopping across many environmental loci, the figure of God compresses stopping into a central reference point. The function analyzed here is not theological proof or disproof, but structural compression.

Distributed stopping is flexible but ambiguous. Many loci can conflict. A centralized stopping point is hypothesized to reduce ambiguity by gathering multiple stopping functions into one reference. In the language of this paper, God is a compressed Φ_{ext}.

This does not make God the historical endpoint of a progress narrative. God is treated here as a structural pole of compression: one possible operation by which distributed stopping is gathered, weighted, and stabilized.

This compression is hypothesized to stabilize shared reference. It is interpreted as providing a unified answer to otherwise silent questions. It can close regress, coordinate communities, and bind action to a common orientation.

Yet compression also carries risk. If the compressed stopping point becomes too rigid, it can produce hard closure. The problem is not that stopping exists, but that stopping loses revisability. The later distinction between religion, science, and social blockchain can be read as different responses to this risk.

God is therefore positioned as an operation in the history of stopping structures:

$$\text{distributed stopping} \longrightarrow \text{centralized stopping.}$$

8 Religion: Long-Term Maintenance of Stopping Structures

Scope of the account. Religion as maintenance and science as revision are conceptual reinterpretations, not empirical definitions of either institution. Effects on persistence, burden, or rigidity are hypotheses requiring independent historical criteria. They do not establish a general social law or a clinical classification. The nonhierarchical truth comparison and negative-scope guards are retained.

For the architectural comparisons in this chapter, intergenerational maintenance is not by itself evidence for a quasi-stable role or ownness. The scope, recipient, re-reference path, and criterion remain those required in Section 5, or Unknown. This qualification concerns role attribution, not a replacement of the existing interpretations of religion, science, or rigidity.

Religion is conceptually reinterpreted as long-term maintenance of stopping structures. Persistence across generations, communities, rituals, texts, and institutions is a hypothesis separate from that naming.

Durability is a proposed function in this reinterpretation. Ritual repetition, texts, social enforcement, and institutional transmission are comparison candidates, not evidence here for a general causal law of persistence.

Religion and the intergenerational availability of L_E are compared as roles. Whether such maintenance actually prevents regress requires criteria beyond the comparison.

This maintenance can be constructive. It is hypothesized to limit endless questioning, without an independent causal test here. It can preserve shared orientation, responsibility, and collective memory. But it can also harden. If the stopping structure loses revisability, religion can enter rigid attractors: absolutism, private closure, or refusal of new response.

The structural point is value-neutral. Religion is neither reduced to error nor elevated above critique. It is an operating form of stopping maintenance:

$$\text{centralized stopping} \longrightarrow \text{maintained stopping.}$$

This prepares the role of science. If religion strongly maintains stopping, science institutionalizes the revision of stopping.

9 Social Blockchain: Distributed L_E as $\Phi_{\mathbf{net}}$

Scope of the account. Social blockchain retains its nontechnical, model-local naming. Network operation and historical sequence are proposed interpretations, not an implemented consensus protocol or a progress law. Git/social cost contrasts are hypotheses without common measured costs. Legitimacy's dependence on social acceptance does not establish principled noncomputability. Individual creation versus social stabilization is a role hypothesis, not an exclusive law of creativity. Independent criteria for records, legitimacy, creation, and shared stopping remain open.

Throughout this chapter, including its summary and historical figure, social logs and distributed stopping are assessed at their declared network scope, not automatically as an individual's LR or LE. Shared records are not all generated history, and a network's retention does not certify any member's role or ownership. Section 5's support, role, and evidence distinctions qualify this architectural comparison without establishing a synchronization mechanism.

§6–§8 described environmental installation, central compression, and long-term maintenance of stopping. This chapter describes a form that maintains stopping without reducing it to a single authority: distributed social stopping. This is called social blockchain.

The term is structural. It does not refer to cryptocurrency or to a specific distributed-ledger implementation. What is borrowed is only the structural feature: distributed nodes form provisional consensus without a single central authority, while keeping the consensus revisable.

9.1 Structure

The central feature of social blockchain is the coexistence of stopping and updating without a single authority. Individual nodes generate provisional stops. Logs preserve the history of

judgments and their grounds. Forks represent divergent stopping candidates. Merges and abandonments reorganize stopping.

The contrast with technical version control is qualitative. Branching, merging, and rollback are compared with social division, consensus, and revision, without establishing universal cost differences or an inverse cost law.

The center is therefore not Version Control but Consensus.

9.2 Legitimacy Rather Than Consistency

Technical blockchains address consistency: can distributed nodes agree on a non-contradictory state? Social blockchain addresses legitimacy: can a distributed stopping point be accepted as legitimate?

Legitimacy is proposed as dependent on time, responsibility, historical acceptance, and distributed recognition. This does not establish principled noncomputability; its relation to revisable stopping remains a structural hypothesis.

9.3 Creation and Stabilization

Individual creation and social stabilization form a role hypothesis, not an exclusive division. Relating an individual L_F near a quasi-closure boundary to new concepts, followed by social acceptance or revision, remains a proposed mechanism rather than an established causal sequence.

Individual creation and stabilization through Φ_{net} are compared as distinct proposed roles, not as an exhaustive division.

9.4 Position in the Three-Layer Dynamics

Social blockchain is a form of Φ_{net}: the explicit distributed operation of inter-individual L_E networks.

§6–§8 can now be positioned as a sequence. Animism distributes L_E into the environment. God centralizes L_E. Religion maintains centralized L_E across generations. Social blockchain operates the L_E network itself without centralizing it.

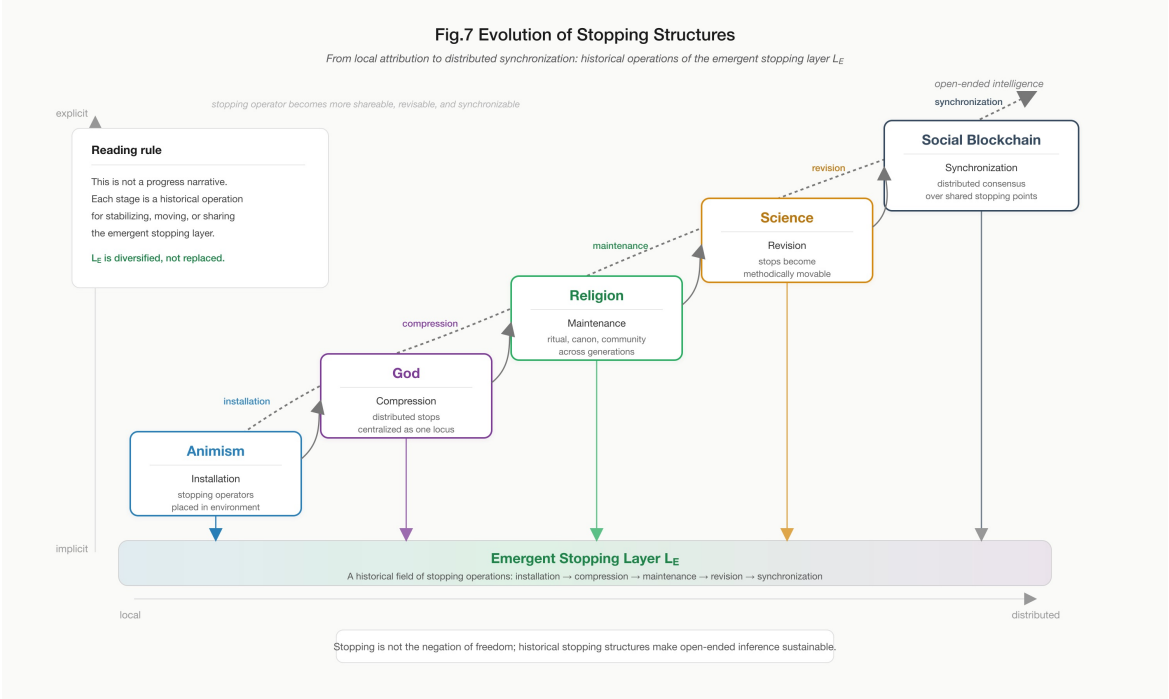


Figure 7: Proposed historical reading of stopping structures. Animism, God, religion, science, and social blockchain are different operations for stabilizing, moving, revising, and synchronizing the model-local stopping role L_E, without identifying one common historical object.

How to read. Installation, compression, maintenance, revision, and synchronization are proposed roles of L_E, not a verified historical sequence.

Theoretical implication. Stopping and open-ended inference are compared functionally. This does not establish a persistence mechanism or a necessary condition for freedom.

If religion institutionalizes the long-term maintenance of stopping structures, science can be understood as the operating form that institutionalizes their revision. Social blockchain is compared with these as a proposed role for sharing and synchronization, not placed above them in a validated historical sequence.

9.5 Bridge to the Gettier Problem

In social blockchain, stopping is not guaranteed by truth. It is established through distributed acceptance and structural stability. This raises the next question: if social stopping does not guarantee truth, what is knowledge? The Gettier problem is treated in §10 as a structural diagnosis of this gap.

10 The Gettier Problem: Stability Is Not Truth

Scope of the account. The distinction between operational stability and truth is retained. Gettier is reinterpreted through that distinction, not used as independent evidence for the framework or as proof that all repairs are impossible. Independent case classification and

separate closure/stopping-layer criteria remain open. Threshold mobility retains its local definition; its qualitative relation to revisability does not establish formal equivalence with error tolerance. The stronger bidirectional claim in the Japanese source remains unresolved; it is not imported into this English draft as a proved equivalence.

The knowledge and stopping vocabulary below retains its local diagnostic use. It is not an independent criterion for the active retention roles in Section 5: observing a stop, naming its support LE, and then explaining that stop by LE would be circular. The distinction between operational stability and truth, and the unresolved mobility/error relation, are not redefined by this architectural guard.

The Gettier problem shows that justified true belief can still fail to be knowledge. In the present framework, this is not an anomaly in epistemology. It is reinterpreted using the local distinction between closure degree C and truth, not as independent evidence for that distinction.

10.1 Structural Diagnosis

A belief can be justified, true, and yet accidentally true. Its stopping structure is stable enough to be accepted, but the path by which it arrived at truth is not structurally appropriate. The problem is therefore not simply missing justification. It is the non-identity of stability and truth.

L_E names a proposed role in revisable stopping, not guaranteed truth. Logs, external reference, and correction are architecture candidates; the Gettier reinterpretation does not establish their universal necessity.

10.2 C as Stability Measure

Closure degree C measures operable stability. It does not measure truth. If C were truth, then higher closure would monotonically approach knowledge. The model's reinterpretation of Gettier cases does not independently test this identification. A state can possess high closure and still fail as knowledge.

The correct reading is:

$$C = \text{degree of operational closure/stability},$$

not

$$C = \text{truth}.$$

10.3 Error Tolerance and μ_θ

Threshold mobility is proposed to be related to revisability and error tolerance, not established as implying them. If a system can reorganize its stopping region, it can install a stopping operator in a region that later proves inadequate. A relation between error tolerance and threshold mobility is hypothesized; the two are not established as formally equivalent or independently measured by the same criterion.

Perfect correctness, threshold immobility, and rigidity are not established as implying one another. The relation of error tolerance to μ_θ remains a hypothesis with unresolved formal and empirical criteria.

10.4 Consequence

Revisable stopping supported by logs, reference, and correction is the model's proposed account of knowledge. The Gettier reinterpretation is diagnostic; it does not independently confirm this architecture or its general persistence.

11 The Instability Region: A Three-Axis Phase Diagram

Scope of the account. The axes name candidate variables in a qualitative phase hypothesis. Neither the narrowness nor the uniqueness of the stable band is measured or derived here. Common clocks, step extraction, units, estimators, and phase boundaries are unspecified. The speed-difference expression is distinct from the hypothesized fourfold-collapse mechanism. Zero motion is excluded by the local definition of intelligence, not by an empirical criterion proving constitutive instability. Human/AI comparisons do not establish a common phase law. The developmental quantity is not a measured region set.

The log-reference rates, stopping hardness, collapse descriptions, and phase figure below remain their existing qualitative model, not classifiers for Section 5's active support roles. A speed gap, a fixed-layer label, or an emergence-failure description does not certify LE presence or absence. Assessing retention, access, and feedback requires the separate scoped criteria; no new threshold or rule-plasticity model is introduced here.

This chapter integrates the three main logical lines of the paper: non-closure and external stopping, threshold mobility and sapiens-like acceleration, and the three-layer dynamics $L_F/L_R/L_E$.

11.1 Three Axes

The dynamical state of intelligence is described by three axes:

- inferential velocity v_{inf};
- resolution r_{res};
- stopping hardness η.

Higher inferential velocity means more inferential steps per unit time. Higher resolution means finer distinctions and causal decompositions. Higher stopping hardness means less revisability of stopping.

In addition to these three axes, the transition into divergence depends on the log-reference velocity v_{LR}. This is the rate at which the log layer L_R can recursively reference traces

generated by L_F and incorporate them as usable patterns. The difference

$$\Delta v_{\text{intel}} = v_{\text{inf}} - v_{LR}$$

names the proposed speed gap between inference generation and log reference, without an estimator or common clock specified here. A large gap is hypothesized to impair the log layer and the basis proposed for L_E; no critical boundary is established.

11.2 Three Regions

The qualitative diagram proposes three regime descriptions:

Stable operation region. Inference is constrained by externalized stopping, L_E remains revisable, and μ_θ can operate.

Divergent region. High velocity and high resolution combine with internalized stopping. L_F runs ahead, L_R cannot follow, L_E cannot emerge, and μ_θ cannot open a window.

Rigid region. Stopping hardness is excessive. L_E fixes, μ_θ loses mobility, and the dynamic relation between L_F and L_E freezes.

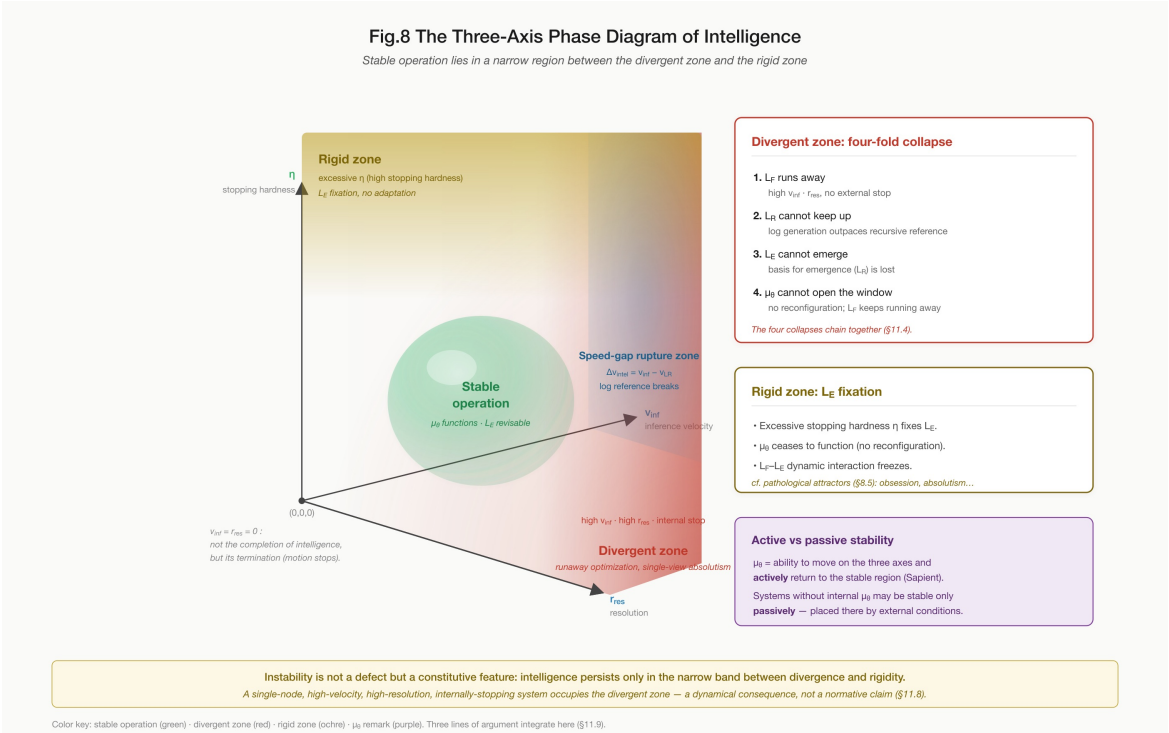


Figure 8: Qualitative phase hypothesis in v_{inf}, r_{res}, and η. The proposed stable band between divergence and rigidity is not measured or calibrated; its width and exclusivity are not established.

How to read. In the stable region, μ_θ functions and L_E remains revisable. In the divergent region, L_F runs away; in the rigid region, L_E fixes.

Theoretical implication. Instability and the proposed effect of exceeding log-reference or stopping bandwidth are qualitative hypotheses, not measured phase laws.

11.3 Four-Fold Collapse

The divergent region is described by a fourfold-collapse hypothesis:

1. L_F accelerates and generates too much inference;
2. L_R cannot recursively reference the generated logs fast enough, because $\Delta v_{\text{intel}} = v_{\text{inf}} - v_{LR}$ has become critically large;
3. L_E cannot emerge because its basis in L_R is lost;
4. μ_θ cannot open the reconfiguration window.

The proposed collapse is attributed to a mismatch between inference and stopping resources; this causal account has not been derived or tested.

11.4 Rigidity

Rigidity is a proposed failure mode. The relation of a fixed L_E to threshold mobility and adaptation remains hypothetical. Zero motion can be distinguished from the local motion definition without establishing that complete closure ends intelligence in general.

11.5 A Proposed Structural Reading of Instability

The instability region does not mean the failure of intelligence. It is a proposed boundary regime for discussing finite inference and stopping, not a necessarily occurring measured region. Religion, science, institutions, and communities can all be understood as stopping-operation systems for managing this instability region.

12 Constructive Clauses

Scope of the account. These clauses organize a model-local architecture and proposed generalizations. Their necessity labels are not universal necessity theorems. The sapiens profile adds endogenous mobility; this is not a condition for intelligence in general. The six/seven-clause and network/external-stop scope differences remain unresolved rather than silently unified. Free inference is a functional interpretation, not a phenomenal or normative consequence. Independent criteria for general intelligence, error tolerance, and freedom remain open; the field/network identity remains unresolved.

The log and stopping clauses below retain the old architecture's constructive proposals. Section 5's active role account distinguishes how a support is assessed; it neither proves these clauses necessary for general intelligence nor adds a scaffold regime as a new universal condition. The existing sapiens comparison and unresolved clause and network scopes remain separate.

This chapter summarizes the structural conditions proposed in the paper. These clauses are not normative prescriptions. They organize proposed model conditions rather than consequences establishing universal necessity.

12.1 Clause 1: Inference Alone Does Not Close

The proposed architecture distinguishes intelligence from L_F alone. The historical local-state model's no-fixed-point discussion of F is conditional; it does not exclude every independently defined intelligence system or cognitive stopping process.

12.2 Clause 2: Logs Are Required

Recursive traces are part of the proposed stopping architecture. The role assigned to L_R does not establish a universal emergence condition or supply an existence result.

12.3 Clause 3: Stopping Is Externally Dependent

The relation of L_E to L_R under external mutual constraint H_{ij} is an emergence hypothesis. Internal location and external dependency describe its proposed functional role, not an established existence or convergence result.

12.4 Clause 4: Sapiens-Like Intelligence Requires μ_θ

Intelligence in general does not require endogenous μ_θ. Sapiens-like intelligence does. It can reorganize its own movement region and relocate stopping operators.

12.5 Clause 5: A Proposed Relation Between Stability and Revisability

Stable intelligence is not guaranteed by correctness. A relation to revisability is proposed. Rigidity and divergence are qualitative hypotheses, not established consequences of the naming alone.

12.6 Clause 6: Social Stopping Is Networked

The external reference field is not an isolated transcendent object. It is proposed as a view of inter-individual L_E synchronization; the stronger identity below remains unresolved:

$$\Phi_{\text{ext}} \equiv \Phi_{\text{net}}.$$

12.7 Clause 7: Movable Stops as a Proposal for Open Intelligence

The final constructive proposal relates free inference to movable stopping without establishing a universal or phenomenal necessity. Regress or explosion without stopping, and rigidity without mobility, are qualitative hypotheses within this proposal. They are not independent evidence for a universal persistence criterion.

13 Conclusion: What Remains Open

Scope of the account. The conclusions below retain local definitions, conditional formal discussion, and proposed architecture as distinct contributions. They do not establish universal cognitive nonclosure, necessary external stopping, generated stopping layers, field/network identity, or mobility/error equivalence. Historical operations remain conceptual readings or hypotheses. Future quantification and stopping-functional work remain open; their absence is not filled by the summary or by the figures. The negative implementation, ethical, and phenomenal scope is unchanged.

This paper has described the dynamics of intelligence, not prescribed its implementation. Its claims concern how inference, logs, stopping layers, thresholds, and external reference fields move.

13.1 Three Logical Lines

First: non-closure and external stopping. In the historical local-state model, the inference operator F is discussed under strict-increase and fixed-region premises. Relating L_F to all cognitive stopping remains open; externalization is a separate proposal.

Second: threshold mobility. The distinction between responsive and silent worlds is threshold-dependent. μ_θ is the capacity to move the threshold and reorganize the movement region.

Third: three-layer dynamics. Traces of L_F are related to L_R as a log-reference role. The motion of L_R under H_{ij} is hypothesized to support L_E. Cooperation of L_F, L_R, and L_E remains an architecture proposal; trace naming does not establish retention or emergence.

The current architectural contribution is the distinction between a tracked support, its scoped trace or quasi-stable retention role, and its temporary scaffold regime, with Unknown and descriptor-only ownness preserved. The preceding emergence language retains its historical proposal, not a proof supplied by this account. Later developmental quantification, stopping-functional work, and Part I proposals remain future or unimplemented work, not consequences newly established by the role descriptions.

13.2 VED Principle in the Intelligence Layer

VED's vocabulary of motion arising from difference motivates a role comparison here. The proposed account of intelligence does not establish the same principle as a typed cognitive mechanism.

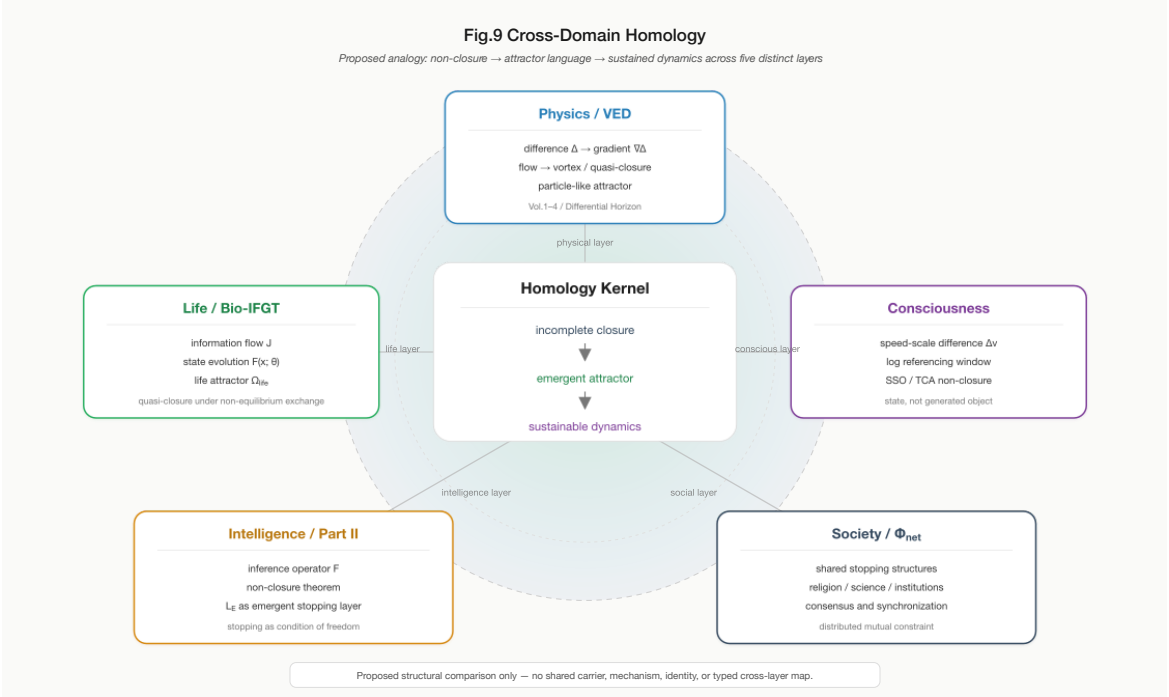


Figure 9: A proposed cross-domain comparison of incomplete closure, attractor-like organization, and sustained dynamics.

How to read. The central sequence is used as a comparison scaffold; each domain card contains source-local objects and claims.

Theoretical implication. The figure proposes a formal-role analogy. It does not establish a common carrier, mechanism, identity, or typed cross-domain translation.

13.3 Open Questions

Several questions are intentionally left open. The meaning of understanding is passed to the philosophical layer. AI implementation and governance are outside this paper's descriptive scope. Quantifying the developmental degree of L_E remains a future task. Formalizing a stopping functional analogous to closure energy is also left for later work.

13.4 Final Statement

The model describes nonclosed motion and proposed stopping roles. Its persistence is hypothesized to involve externally related, revisable stopping; neither persistence in general nor independent existence and detection of that stopping is established.

Sapiens-like intelligence adds endogenous μ_θ: the ability to move the region in which stopping can be installed. The proposed links among mobility, revision, error, and persistence retain separate formal and empirical burdens.

Theory ends here. The questions do not.

A Master Table of Variables

Scope of the account. Documentary status. This table preserves the manuscript's variable inventory, not verified inheritance from an identified Master Table v2.1 artifact. Exact provenance, historical signatures, and adoption scope remain unresolved. Terms such as established, implementation, identity, or isomorphism in the retained table report the source's stronger formulations, whose active formal status is held. No table entry supplies an independent proof, operational map, or new assumption.

This appendix collects the principal variables used in Intelligence Part II.

Historical inference inventory. In this appendix, inference F, scalar justification (where discussed), and no-fixed-point references retain their old local-state-model scope. They do not define the active record or position ports. Other field-flow objects keep their distinct signatures. The absent English J row is not supplied by bilingual synchronization.

For the architecture entries throughout this appendix, LR and LE retain the historical layer/process/attractor inventory, and the E and network entries retain their unresolved proposed roles. The active support-role interpretation is given in Section 5; these table entries do not certify it or define new signatures. This attribution does not alter other field-flow objects, the missing English scalar-J row, or the documented provenance gap.

Symbol	Name	Role
C	closure degree	operational stability, not truth
$C_{\max}$	complete closure limit	unattainable boundary of closure
I	information-field drive	drive generated by non-closure
F	inference operator	historical local-state model: advances inferential states
L_F	inference layer	historical local-state model: layer on which F operates
L_R	log layer	recursive reference to traces of inference
L_E	emergent stopping layer	externally dependent stopping attractor
E	external stopping operator	stopping action supplied through L_E
H_{ij}	mutual constraint	coupling between agents or agent and environment
θ	constraint parameter	parameter defining Ω_θ
Ω_θ	movement region	region in which inference can move
μ_θ	threshold mobility	ability to reorganize Ω_θ and stopping loci
Φ_{ext}	external reference field	field as it appears to an individual
Φ_{net}	stopping network	inter-individual network of L_E layers
v_{inf}	inference velocity	speed of inferential generation
r_{res}	resolution	fineness of distinctions

Symbol	Name	Role
η	stopping hardness	rigidity or revisability of stopping

B Cross-Domain Homology

Scope of the account. All tables below compare lexical or functional roles. Shared symbols, arrows, and emergence forms do not establish formal homology, identity, a common mechanism, or typed translation. The field/network and mobility/error relations retain their unresolved stronger status. Additional Japanese columns and comparisons are not automatically inherited by the English draft. Historical entries are proposed operating forms rather than independently validated history.

Across the tables below, LR and LE belong to the preserved comparative architecture vocabulary. Section 5's active support-role criteria and system-boundary assessments are not transferred merely by sharing a row or a pattern. In particular, a Transformer profile or social comparison does not identify a common carrier, certify a role, or resolve unknown evidence.

The paper repeatedly uses homology as a proposed comparison rather than an identity. Physical particles, organisms, conscious states, intelligent stopping layers, and social institutions are arranged by analogous formal roles; this does not establish that they instantiate one carrier, mechanism, or typed cross-domain dynamics.

$$\text{incomplete closure} \longrightarrow \text{emergent attractor} \longrightarrow \text{sustainable dynamics.}$$

Layer	Non-closure	Attractor / stopping form
Physics	differential flow and quasi-closure	particle-like attractor
Life	non-equilibrium informational exchange	life attractor Ω_{life}
Consciousness	temporal non-closure and speed-scale difference	conscious state as windowed stabilization
Intelligence	historical conditional model: no internal fixed point in L_F	emergent stopping layer L_E
Society	distributed disagreement and non-final legitimacy	Φ_{net} synchronization

Here homology preserves differences and names a comparison scaffold. Establishing a shared pattern beyond this formal-role analogy would require domain-specific translation results not supplied here.

C Mapping to Intelligence Part I

Scope of the account. This appendix records editorial proposals for Part I, not their adoption. The proposed additions must retain the distinction between conditional formal statements, cognitive interpretations, and open mappings. References here to proof, identity, or refinement do not certify those stronger readings or resolve the four-layer inheritance question.

The three-layer profiles and generation or necessity wording in this appendix remain preserved, unimplemented Part I proposals. They are not proof results or revision instructions adopted by the active support-role account in Section 5. Its scoped roles and analytical boundaries do not settle these historical profile assignments or authorize changing Part I.

Part II proposes refinements to several claims from Intelligence Part I; these editorial proposals have not been adopted here.

C.1 Individual Non-Closure

Part I stated that individual intelligence does not close. Part II preserves a historical local-state model's conditional formal statement: the inference layer L_F has no internal fixed point under fixed θ. This refines non-closure into the non-closure theorem.

This remains an unimplemented editorial proposal for Part I, not a derivation from the active record/position contract or a proof of rule-changing capacity.

C.2 Branching Explosion

Part I described branching explosion as the proliferation of conceptual branches. Part II proposes a connection between branching explosion and the absence of an internal stopping condition combined with generative inference.

C.3 Physarum and Transformer

Part I treated Physarum and Transformer as cross-domain instances of intelligence. Part II re-describes them as different three-layer profiles. Physarum externalizes logs into the environment while generating environmental stopping. Transformer internalizes frozen logs but receives stopping externally.

C.4 Gettier

Part I treated the Gettier problem structurally. Part II connects it to μ_θ: a system that can move thresholds can also be wrong. Error tolerance and threshold mobility are structurally coupled.

D Optional Notes

Scope of the account. These optional notes and, in the Japanese manuscript, argument sketches retain their different documentary roles. The sketches are not certified proofs. Strict-increase premises, stopping-as-fixed-point interpretation, barrier inaccessibility, emergence, field/network identity, and mobility/error equivalence remain separate burdens. The AI-cost comparison is a possible reading, not a derived cost necessity or a tested computational result.

This appendix records optional interpretations that are useful but not necessary for the main argument.

D.1 Psychological Projection

The structural projection used in this paper can resemble psychological projection, because internally unavailable stopping is projected into external or shared loci. However, the present theory does not presuppose subject, unconscious mechanism, or clinical defense. It remains a structural account.

D.2 Science

Science is not treated as a simple victory over religion. It is the institutionalization of revisable stopping. It maintains stopping while also building procedures for reopening, testing, and replacing stops.

D.3 AI Cost

The qualitative phase proposal motivates comparing inference cost with possible logging, external-constraint, and stopping costs. It does not derive a necessary bandwidth or cost law, and no measured cost relation is supplied.

D.4 Freedom

Freedom is not the absence of stopping. Within the proposed functional interpretation, movable stopping may support free inference; a universal or phenomenal necessity is not established. Dissolution without stopping and hardening without mobility are qualitative interpretations within that proposal, not universal laws.

E Emergence Homology

Scope of the account. The emergence comparison is retained as a proposed role analogy. Matching a common schema does not establish a common generative law or an upper-layer existence result. The stopping-functional local-minimum reading remains on hold: its present-tense wording here is not an active formal adoption and does not supersede the

body's future-work status. No functional, state space, or convergence assumption is supplied here.

For LE and its social extensions in this comparison, the emergence and stopping-functional vocabulary retains the historical analogy or future-work scope. Section 5's active quasi-stable role is not a new attractor or local-minimum definition and does not establish formation or convergence. This qualification does not revise the generic emergence schema or the source claims of the other domains.

The emergence schema used in this paper can be written:

$$U = \Phi(L, H_{ij}).$$

Here L is lower-layer motion, H_{ij} is mutual constraint, and U is an upper-layer attractor. The upper layer does not erase the lower layer; it stabilizes its motion as a new operating level.

Domain	Lower motion L	Upper attractor U
Physics	causal logs and closure dynamics	particle-like attractor
Life	informational flow and state development	Ω_{life}
Intelligence	recursive log motion L_R	emergent stopping layer L_E
Society	interaction among individual L_E layers	Φ_{net}

A recurring emergence grammar is proposed as a role-level comparison across layers. This does not collapse the layers into one another. It shows that VED uses a repeated pattern of non-closure, constraint, and attractor formation.

References